

High-antioxidant potatoes: acute in vivo antioxidant source and hypotensive agent in humans after supplementation to hypertensive subjects.

Potatoes have the highest daily per capita consumption of all vegetables in the U.S. diet. Pigmented potatoes contain high concentrations of antioxidants, including phenolic acids, anthocyanins, and carotenoids. In a single-dose study six to eight microwaved potatoes with skins or a comparable amount of refined starch as cooked biscuits was given to eight normal fasting subjects; repeated samples of blood were taken over an 8 h period. Plasma antioxidant capacity was measured by ferric reducing antioxidant power (FRAP). A 24 h urine was taken before and after each regimen. Urine antioxidant capacity due to polyphenol was measured by Folin reagent after correction for nonphenolic interferences with a solid phase (Polyclar) procedure. Potato caused an increase in plasma and urine antioxidant capacity, whereas refined potato starch caused a decrease in both; that is, it acted as a pro-oxidant. In a crossover study 18 hypertensive subjects with an average BMI of 29 were given either six to eight small microwaved purple potatoes twice daily or no potatoes for 4 weeks and then given the other regimen for another 4 weeks. There was no significant effect of potato on fasting plasma glucose, lipids, or HbA1c. There was no significant body weight increase. Diastolic blood pressure significantly decreased 4.3%, a 4 mm reduction. Systolic blood pressure decreased 3.5%, a 5 mm reduction. This blood pressure drop occurred despite the fact that 14 of 18 subjects were taking antihypertensive drugs. This is the first study to investigate the effect of potatoes on blood pressure. Thus, purple potatoes are an effective hypotensive agent and lower the risk of heart disease and stroke in hypertensive subjects without weight gain.